

Unit-2 : Disaster Management Cycle and Framework

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Topics to be covered

- ❑ Disaster Management Cycle
- ❑ Paradigm Shift in disaster management
- ❑ Various Frameworks like Yokohama Strategy, Hyogo Framework, Sendai Framework etc.

Disaster Management – How?

Now that we have a clear idea of how bad disasters can be, we obviously need to manage it. This is how disaster management came into being.

The following definition of Disaster Management has been taken from the Disaster Management Act (DM Act), 2003.

Disaster Management means a continuous and integrated process of planning and implementation of measures for:

1. Mitigating the risk of disasters;
2. Mitigating the severity of disasters;
3. Capacity - building;
4. Emergency preparedness;
5. Assessing the effects of disasters;
6. Providing emergency relief and rescue; and
7. Post-disaster rehabilitation and reconstruction

Managing disasters: Disaster Management Cycle

What is Disaster management cycle?

- The disaster management cycle is a framework that defines the stages of a disaster.
- It can be used by both organizations and individuals to prepare for and respond to disasters of every kind, including natural disasters, technological disasters, and human-made disasters.
- It allows professionals to identify potential hazards, assess the risks, and develop plans to prevent, mitigate, and respond to them.
- Unprecedented disasters can strike at any time. But, using the disaster management cycle as a guide, professionals can create plans that can help prevent the worst effects and lay the groundwork for a more effective recovery.
- For instance, during a flood, a [disaster plan](#) may involve enabling communications, blocking off roads, organizing evacuations, or enabling victims to retrieve valuable items from flooded areas.

- The disaster management cycle includes recovery from the disaster, both for the affected individuals and their property. The recovery process may involve rebuilding, providing medical attention, and creating a safe environment, and can also extend to an area's infrastructure, economy, population, and ecosystem.
- Different disasters require different methods to ensure the highest chance of prevention and an expedient return to everyday life for everyone involved if the disaster becomes unavoidable. Trained professionals are key in this endeavor, as they can best determine the required level and type of assistance. For example, in a wildfire, disaster management may involve fighting the blaze, whereas, during a hurricane, the priority may be to search for survivors.

Disaster management cycle

Disaster management typically is broken down into four stages:

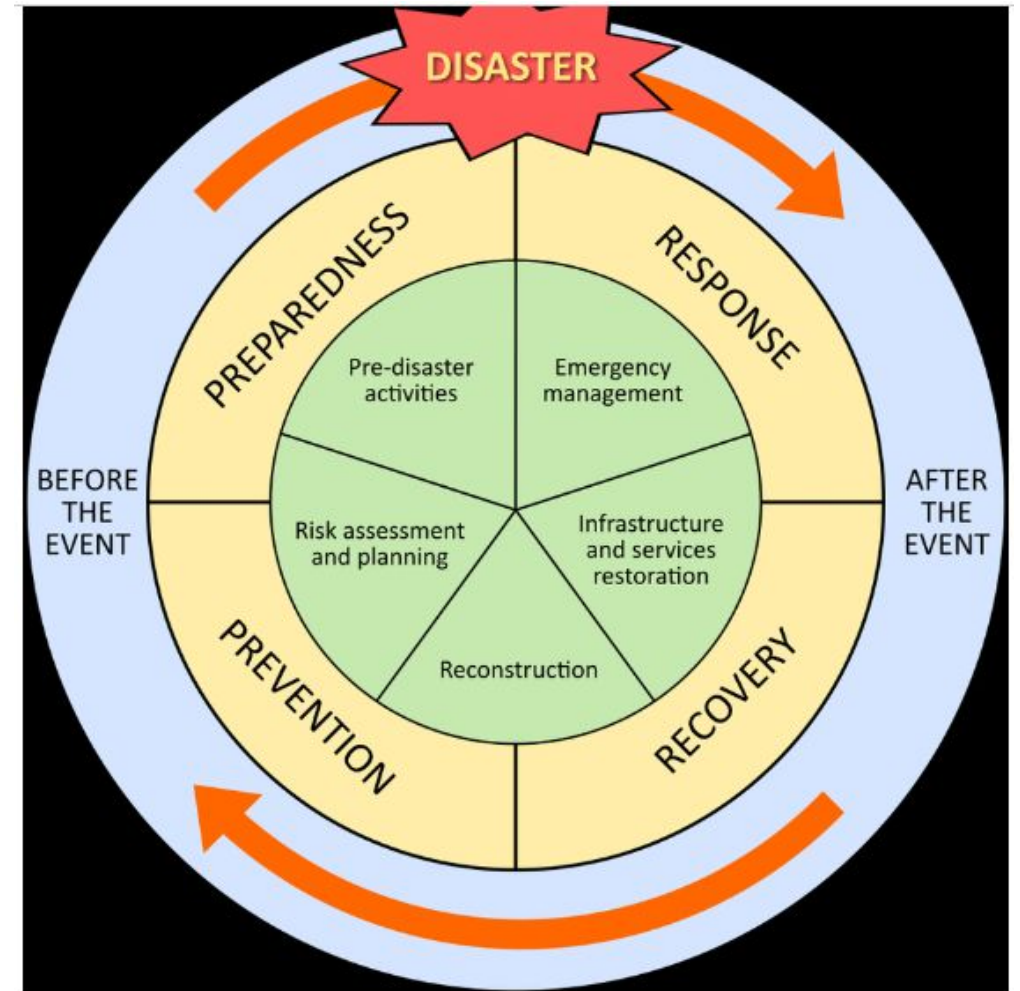
1. Prevention
2. Preparedness
3. Response
4. Recovery.

- **Prevention**

Measures taken to avoid the occurrence of a disaster

- **Preparedness**

State of readiness which enables stakeholders to mobilize, organize and provide relief to deal with an impending or actual disaster or the effect of a disaster

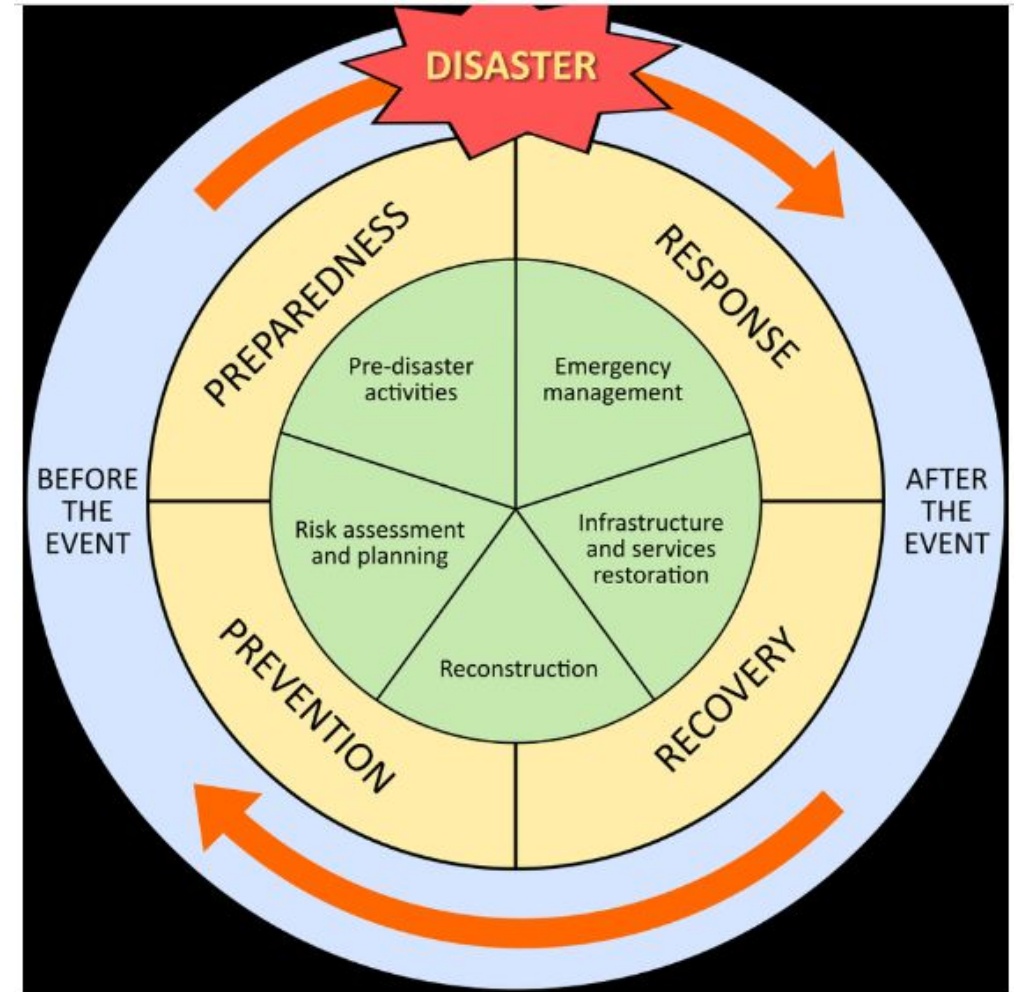


- **Response**

Measures taken during or immediate after a declaration of disaster to diminish, or alleviate any suffering, pain, injury or distress or hardship caused on account of the disaster

- **Recovery**

The actions taken by the community and all the government and the non-government agencies to reset the condition after disaster and minimize the vulnerability of the same kind of disaster in future



From Disaster Management to Disaster Risk Management

We need to manage risks, not just disasters.

- Don't you think disaster management is more of a reactive measure? You are right!
- Taking cognisance of this and powered by the researches that are going on around the world,
- we are slowly shifting towards disaster risk management.
- **Disaster Risk Management (DRM) is primarily the management of disaster risks in a way such that those risks never become a disaster!**
- You may have heard of a term, Disaster Risk Reduction. What is disaster risk reduction?
- **Disaster Risk Reduction (DRR) is all that we do to ensure that we have managed the risks that we know about and be prepared to deal with the ones that we do not know about.**

From Disaster Management to Disaster Risk Management

**Disaster Risk Management (DRM) =
Disaster Risk Reduction (DRR) + Disaster Management
(DM)**

Resilience to disaster: Way forward

Resilience is the ability of a system, community or society exposed to hazards to

- resist,
- absorb,
- accommodate to and
- recover from

the effects of the hazard in a timely and efficient manner, including through the preservation and restoration of its essential basic structures and functions

- **How do we become resilient?** It is a complex net of many elements.
- It is multi-disciplinary, trans-disciplinary.
- Technical science, social science, medical science, environmentalists etc., everyone is involved, directly or indirectly
- **Just to get an idea of how this will work:**
- Hazard -> Early warning -> Information dissemination -> Impact -> Rescue & Relief -> Reconstruction & Rehabilitation -> Building back better (All this is according to DM Cycle and SFDRR)

Food for thought

- The level of preparedness of Japan is usually understood to be very good. This can be attributed to the fact that it witnesses earthquake and typhoon almost every other day. *To attain that level of preparedness, does India need to suffer the same frequency of recurring hazards?*
- Again, it is inherent to a human to think that disaster will not happen to me and thereby he or she lacks in preparedness. *So, should one suffer a disaster to be prepared?*



The Paradigm Shift

Managing Disaster ————— Managing Disaster Risks

The Paradigm Shift



- It started with the United Nations observing the 1990s as the **International Decade for Natural Disaster Reduction (IDNDR)**.
- Aim was to decrease the loss of life, property destruction and social and economic disruption caused by Natural Disasters.

The Paradigm Shift

Frameworks

- ❑ 1st World Conference on ~~Natural Disaster~~, (May 1994, Yokohama, Japan)
- ❑ Adopted the ~~Yokohama Strategy~~ for a Safer World: Guidelines for ~~Natural Disaster~~ Prevention, Preparedness and Mitigation and its Plan of Action.
- ❑ It was the main outcome of the ~~mid-term review of the International Decade of Natural Disaster Reduction (IDNDR)~~ and established 10 principles for its strategy, a plan of action and a follow-up.
- ❑ In Dec. 1999 United Nations International Strategy for Disaster Reduction (UNISDR) was created - successor to the secretariat of the IDNDR

1994

- 1st World Conference on Natural Disaster
- Yokohama Strategy for a Safer World

1999

United Nations International Strategy for Disaster Reduction (UNISDR)

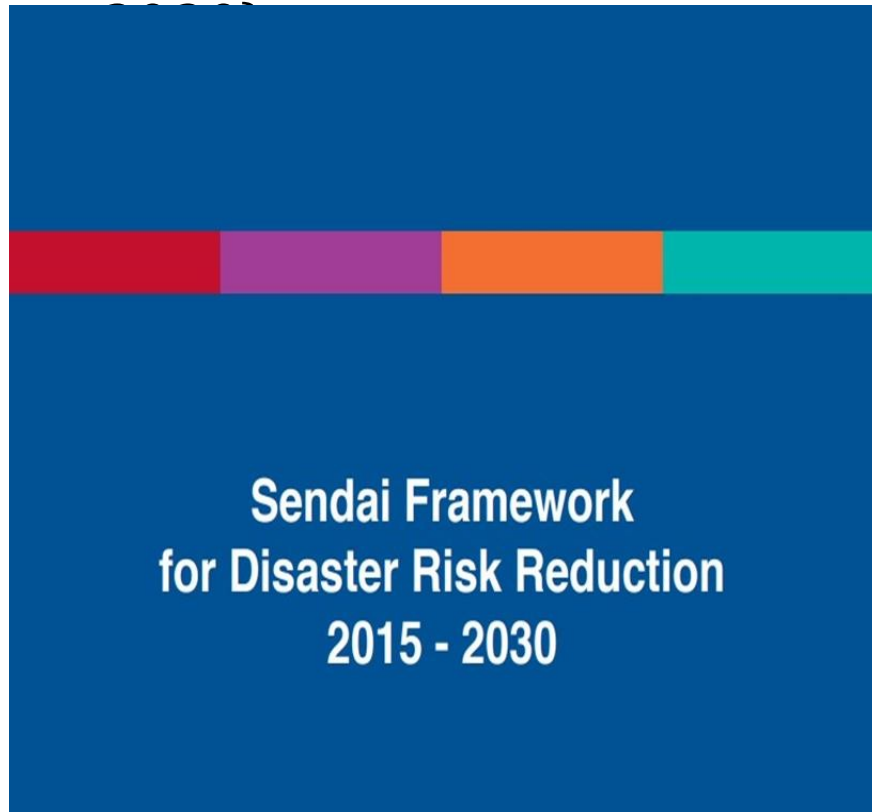
2003 -
2004

UNISDR carried out a review of the Yokohama Strategy and Plan of Action for a Safer World

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- ❑ 2nd World Conference on Disaster Reduction (January 2005, Kobe)
 - ❑ The **Hyogo Framework for Action (HFA) (2005 – 2015)**: Building the Resilience of Nations and Communities to Disasters
 - ❑ Priorities for Action
 1. Making disaster risk reduction a priority;
 2. Improving risk information and early warning;
 3. Building a culture of safety and resilience;
 4. Reducing the risks in key sectors;
 5. Strengthening preparedness for response

Sendai Framework for Disaster Risk Reduction: Latest Framework of DRM

- 3rd World Conference on Disaster Risk Reduction (March 2015, Sendai)
- Adopted the **Sendai Framework for Disaster Risk Reduction (SFDRR)** (2015 – 2030)



Sendai Framework for Disaster Risk Reduction: Latest Framework of DRM

1 OUTCOME

The substantial reduction of disaster risk and losses in lives, livelihoods and health and in the economic, physical, social, cultural and environmental assets of persons, businesses, communities and countries

1 GOAL

Prevent new and reduce existing disaster risk through the implementation of integrated and inclusive economic, structural, legal, social, health, cultural, educational, environmental, technological, political and institutional measures that prevent and reduce hazard exposure and vulnerability to disaster, increase preparedness for response and recovery, and thus strengthen resilience

4 PRIORITIES

Understanding disaster risk

Strengthening disaster risk governance to manage disaster risk

Investing in disaster risk reduction for resilience

Enhancing disaster preparedness for effective response, and to “Build Back Better” in recovery, rehabilitation and reconstruction

7 TARGETS

- ↓ DISASTER MORTALITY BY 2030
- ↓ NUMBER OF AFFECTED PEOPLE BY 2030
- ↓ ECONOMIC LOSS BY 2030
- ↓ INFRASTRUCTURE DAMAGE BY 2030
- ↑ DRR NATIONAL/LOCAL STRATEGIES BY 2020
- ↑ INTERNATIONAL COOPERATION BY 2030
- ↑ EWS AND DR INFORMATION BY 2030

Hyogo Framework 2005-15	Sendai Framework 2015-30
The substantial reduction of disaster losses.	The substantial reduction of disaster risk and losses both.
Minimizing the impacts of disasters.	Efforts to reduce the size of disasters , which would then have an overall effect of reducing its impacts.
An example is risk insurance. Used purely to transfer disaster risks it can result in reduced economic loss, but would not necessarily reduce the size of the disaster.	A focus on disaster risk would use measures such as risk transfer and insurance to also build resilience and knowledge
Focus on the "what" i.e. what must be done in response to those risks.	Focus on the "how" i.e. how are we to prevent or stop the natural hazards (implementation).
Scope include natural, environmental and technological hazards	Scope include natural, environmental, technological, man-made & biological hazards

Key themes of SFDRR 2015-30



- 1. Building Resilience**
- 2. Promoting Local Solution**
- 3. Fostering Inclusion**

Materializing DRR through Sendai Framework

Remember we were talking about Disaster Risk Management (DRM) and Disaster Risk Reduction (DRR)

- Well, it was through the Sendai Framework (SFDRR) that these actually started getting materialised.
- The concepts of understanding disaster risks, managing and reducing them came out of the academics bounds of recommendations and reports.
- All the signatory countries of SFDRR, have started imbibing the principles of risk management and risk reduction in its national and local strategies and policies.
- **India's National Disaster Management Plan is in fact inspired from SFDRR.**

Materializing DRR through Sendai Framework

- Disaster and development are the two faces of the same coin. Do you agree?
- Development, uncontrolled and Sudden/Irregular, exposes us to new hazards, rendering us vulnerable, leading to disasters.
- On the other hand, disasters give us an opportunity to learn from our mistakes and develop in a much more better way!



Materialising DRR through Sendai Framework

But we cannot avoid development? Can we? **Perhaps no!**

So what do we do?

We need to develop in a **sustainable** manner.

Based on this simple principle the United Nations adopted the
Sustainable Development Goals!

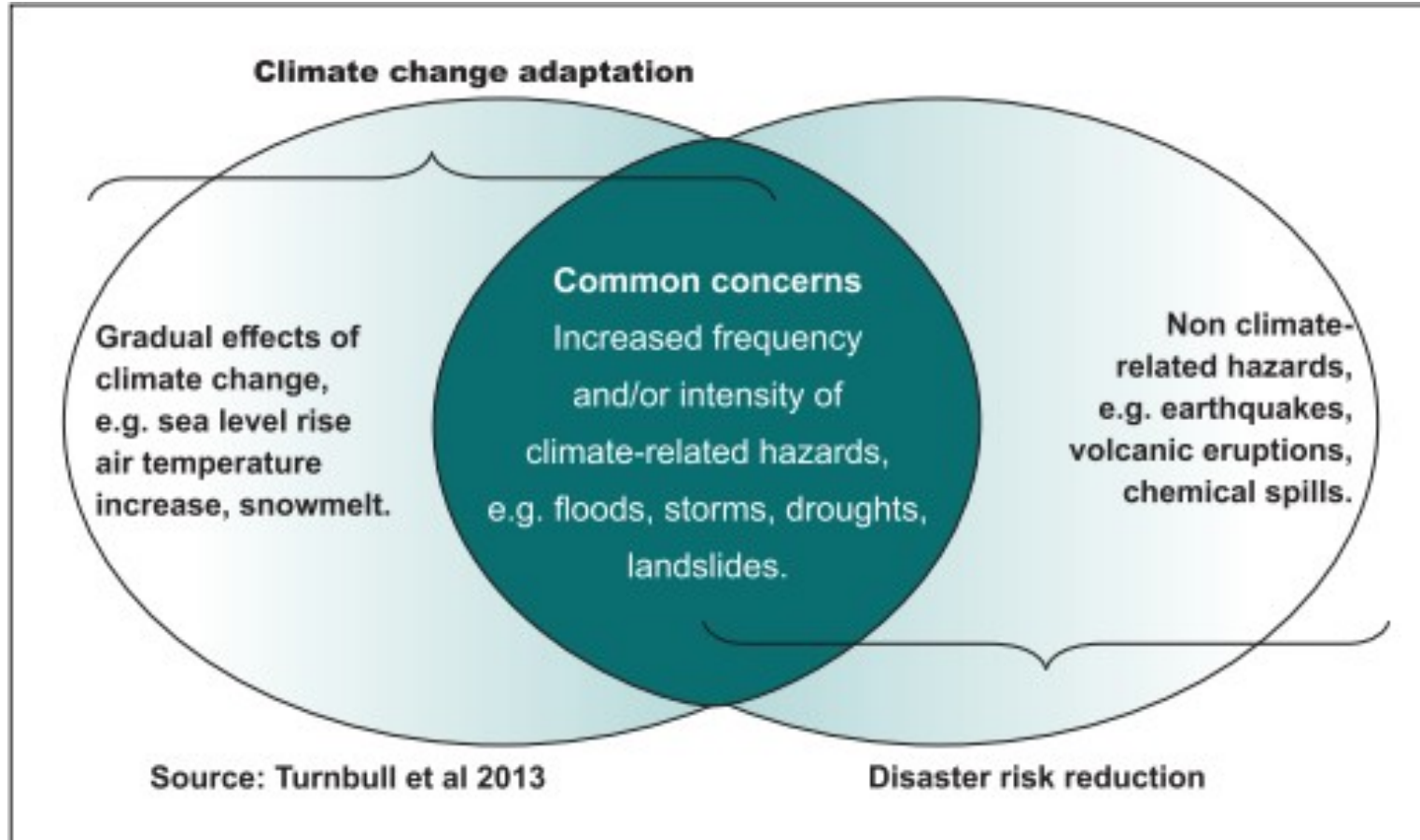
Sustainable Development Goals (SDG)



COP21 Paris Agreement

- The Paris Agreement was adopted on 12 December 2015 at the Twenty-first session of the Conference of the Parties (COP21) to the United Nations Framework Convention on Climate Change (UNFCCC).
- **Goals:**
- A consensus on adopting the long-term goal of keeping the increase in global average temperature to well below 2°C above pre-industrial levels.
- Aim to limit the increase to 1.5°C, since this would significantly reduce risks and the impacts of climate change.
- Accepting the need for global emissions to peak as soon as possible, recognizing that this will take longer for developing countries.
- To undertake rapid reductions of emissions in accordance with the best available science.

Convergence between Disaster Risk Reduction and Climate Change





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, KIGALI ACCORD 2016

Kigali Accord 2016

- The **Montreal Protocol** is an international treaty designed to protect the ozone layer by phasing out the production of numerous substances those are responsible for ozone depletion. It was agreed on 16th September 1987, and entered into force on 1st January 1989. Since then, it has undergone nine revisions last being Kigali 2016.
- In order to abide by Montreal Protocol, it was decided to phase out usage of chlorofluorocarbons (CFCs) and use hydrofluorocarbons (HFCs) as a replacement.
- HFCs are not as harmful as CFCs for ozone layer depletion but they have a thousand times more potential to cause global warming effect than commonly known greenhouse gases like carbon dioxide, methane etc.



Kigali Accord 2016

- The amendment was agreed upon at the twenty-eighth Meeting of the Parties to the Montreal Protocol held on **October 15, 2016, in Kigali. It was decided that:**
- The first group consists of rich and developed economies like USA, UK and EU countries who will start to phase down HFCs by 2019 and reduce it to 15% of 2012 levels by 2036.
- The second group consists of emerging economies like China, Brazil as well as some African countries who will start phase down by 2024 and reduce it to 20% of 2021 levels by 2045.
- The third group consists of developing economies and some of the hottest climatic countries like India, Pakistan, Iran, Saudi Arabia who will start phasing down HFCs by 2028 and reduce it to 15% of 2024-2026 levels till 2047.

Kigali Accord 2016 : Significance

1. It **strengthens the Paris Agreement** which sets an ambitious target of restricting the rise in global temperature below 2° Celsius, as compared to pre-industrial level.
2. Unlike Paris agreement, **it gives clear, concrete and mandatory targets** with fixed timelines to the signatory parties to achieve their targets.
3. It would **prevent the emission of HFCs** equivalent to 70 billion tons of CO₂.

THANK YOU

